



SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

IT OPERATION SUPPORT SYSTEM (ITOSS)

- Project: IT Operations Support System (IOSS)
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1. INTRODUCTION

1.1 Purpose

This document provides a comprehensive and detailed specification of the functional and non-functional requirements for the **IT Operation Support System (ITOSS)**. The system is designed to enable centralized, efficient, and accountable management of all digital banking equipment—including ATMs, POS devices, DOB terminals, computers, printers, and scanners—across the bank's branch network.

The primary objectives of ITOSS are to:

- Monitor ATM uptime and ensure compliance with defined Service Level Agreements (SLAs).
- Digital Channel Management such as POS , and ATM
- Card Production and Distribution Management
- Improve real-time visibility and tracking of IT assets
- Standardize and automate maintenance workflows to reduce resolution times.
- Enforce strict accountability through controlled handover and takeover processes.
- To Manage subscriptions such as Data VPN and Fixes lines
- Provide actionable, management-level reporting and dashboards.
- Ensure a complete, immutable audit trail for all system activities.
- To support & manage all activities in IT Infrastructure department

This SRS serves as the definitive agreement between all stakeholders—including business owners, IT operations, compliance, and the development team—and will be the foundation for system design, development, testing, and acceptance.

1.2 Scope

ITOSS is a secure, web-based enterprise application built for internal use by the bank's IT Operations department. It operates within the bank's trusted internal network.

In-Scope (Major Capabilities):

- **Enterprise Digital Equipment Inventory:** Centralized registry and lifecycle management of all hardware assets.
- **Maintenance Request Lifecycle Management:** End-to-end process from ticket creation to resolution and closure.
- **Custody Management:** Formalized handover and takeover processes for equipment accountability.
- **ATM Performance Monitoring:** Manual recording and automated calculation of uptime, downtime, and related metrics.
- **SLA Compliance Tracking:** Monitoring of maintenance response and resolution times against agreed service levels.
- **Operational Dashboards:** Real-time views of key performance indicators (KPIs) for managers and supervisors.
- **Audit Trail:** Comprehensive logging of all user and system actions for security and compliance.
- **Role-Based Access Control (RBAC):** Granular permissions ensuring users access only authorized functions.
- **Reporting & Analytics:** Generation of standard and ad-hoc reports for operational and management review.

Out-of-Scope:

- Direct low-level control or interfacing with ATM hardware or other devices.
- Real-time automated monitoring of device states (initially; see Future Enhancements).
- External customer-facing functionalities.

1.3 Business Objectives

Establish ITOSS as a centralized platform to control, monitor, and optimize IT assets, channels, and technical operations across all branches and districts.

- Reduce overall ATM and digital channel downtime by **25% within 12 months** of full deployment.
- Improve Mean Time to Repair (MTTR) for critical IT equipment by **20% within the first year** through structured incident tracking and categorized problem management.
- Achieve **100% digital asset lifecycle accountability**, including custody chain, transfer history, and disposal records, across all branches and districts.
- Decrease unplanned digital asset failures by **30%** through enforced preventive maintenance scheduling and compliance monitoring.
- Eliminate **100% of manual spreadsheet-based reporting** within 6 months by automating SLA, inventory, and performance reporting.
- Provide executive management with **real-time dashboards and district-level KPIs** for digital asset utilization, downtime trends, and operational risk indicators.
- Reduce connectivity and subscription waste (VPNs, fixed lines, redundant services) by **15% within the first year** through centralized monitoring.
- Ensure **complete audit trail coverage** for all digital asset movements, user activities, and configuration changes to strengthen governance and regulatory readiness.
- Standardize problem categorization and resolution workflows to reduce recurring incidents by **at least 20% year-over-year**.

1.3.1 Expected Business Impact

- Increased asset accountability
- Reduced downtime and service disruption
- Improved audit readiness
- Lower operational risk
- Better management visibility
- Controlled IT operational costs

ITOSS positions IT operations as a measurable, controlled, and accountable business function rather than a reactive support unit.

1.4 Definitions and Acronyms

Term	Definition
ATM	Automated Teller Machine
POS	Point of Sale Device
DOB	Digital Online Banking Terminal
SLA	Service Level Agreement
MTTR	Mean Time To Repair
MTBF	Mean Time Between Failures
RBAC	Role-Based Access Control
ITOSS	IT Operation Support System
KPI	Key Performance Indicator

2. OVERALL DESCRIPTION

2.1 Product Perspective

ITOSS is a standalone enterprise application residing within the bank's secure internal data center. It will be integrated into the existing IT operational ecosystem.

Architecture Model:

- **Client Layer:** Modern web browsers (Chrome, Edge, and Firefox) accessed via the bank's intranet.
- **Application Layer:** Laravel 12 framework application.
- **Admin Interface:** Filament 4+ for administrative backend operations.
- **Data Layer:** MySQL 8+ relational database.
- **Optional Integrations (Future/Phase 2):**
 - ATM monitoring tool APIs for automated status feeds.
 - Corporate SMS gateway for alert notifications.
 - Internal email server for notifications and report distribution.

2.2 System Overview

ITOSS will serve as the system of record for the bank's branch IT infrastructure, managing the complete lifecycle of equipment, orchestrating maintenance operations, analyzing performance data, and generating compliance reports. A core design principle is the enforcement of full traceability for every action.

2.3 User Classes and Characteristics

User Class	Responsibilities & Characteristics
1. System Administrator	Full system control. Manages user roles, system configurations, backups, and has database-level access.
2. IT Operations Manager	Oversees national/regional performance. Uses dashboards, reviews escalations, approves major actions, and analyzes SLA reports.
3. Operational Manager	Manages a cluster of branches. Assigns tasks to Branch IT Officers, reviews regional uptime metrics and pending requests.
4. Branch User	Front-line personnel. Registers equipment, reports downtime, submits maintenance requests, updates resolution details, and executes handovers.
5. Auditor/Compliance Officer	Read-only access. Examines audit logs, generates compliance reports, and verifies process adherence.
6. System admin Head	The user who supervise the all system administrator in city and upcountry

2.4 Operating Environment

- **Server Environment:** Linux (Ubuntu LTS 22.04 / RHEL 9), PHP 8.2+, Laravel 12, Composer, MySQL 8+.
- **Client Environment:** Modern HTML5-compliant browsers (last two stable versions), accessed exclusively via the bank's internal network.
- **Security Environment:** HTTPS/TLS 1.3 enforced, application behind enterprise firewall, access restricted to specific internal IP ranges.

2.5 Design and Implementation Constraints

- Must adhere to the bank's **IT Security Policy** and **Data Governance Framework**.
- Must integrate with the bank's existing **Active Directory/LDAP** for authentication (if applicable).
- All data must persist within the bank's **on-premise infrastructure**; no cloud storage.
- Must support the bank's standard **database backup and disaster recovery** procedures.
- Must comply with **financial regulatory and internal audit** requirements for data retention and logging.

2.6 Assumptions and Dependencies

- Branch IT personnel will report downtime and maintenance requests **promptly and accurately**.
- **Stable and reliable** network connectivity will be available between branches and the central server.
- A **standardized naming/numbering convention** for Equipment IDs exists or will be established.
- End-users will receive **adequate training** prior to system rollout.

3. SPECIFIC REQUIREMENTS

3.1 Functional Requirements

3.1 Equipment Management Module

- **FR-EM-01: Equipment Registration**
 - The system shall provide a form to register new equipment, capturing: Unique Asset Tag ID, Serial Number, Model, Vendor, Assigned Branch, Purchase Date, Warranty Expiry Date, and Initial Status.

- **FR-EM-02: Equipment Tracking**

- The system shall maintain a complete history log for each asset, including: branch transfers, status changes (e.g., Active -> Under Maintenance), and a running count of maintenance events.

- **FR-EM-03: Equipment Categorization**

- The system shall allow administrators to define and manage equipment types (ATM, POS, DOB, Computer, Printer, Scanner, and Network Device) via a dynamic configuration interface.

3.2 Maintenance Management Module

- **FR-MM-01: Request Creation**

- The system shall auto-generate a unique Ticket ID (e.g., INC-YYYYMMDD-001) and allow a user to create a request by selecting: Equipment, Issue Category, Severity (Low, Medium, High, Critical), and Description.

- **FR-MM-02: Assignment & Workflow**

- The system shall allow supervisors to assign a ticket to a technician, automatically triggering status changes and starting SLA timers. The ticket shall progress through statuses: Open -> Assigned -> In Progress -> (Optional: Escalated) -> Resolved -> Closed.

- **FR-MM-03: Resolution Tracking**

- Upon resolution, the assigned technician shall be required to input: Root Cause, Actions Taken, Parts Used (with serials), and actual Resolution Date/Time.

- **FR-MM-04: SLA Management**

- The system shall allow configuration of target resolution times (e.g., Critical: 2 hours, High: 4 hours). It shall visually flag breaches, calculate MTTR, and provide SLA adherence % in reports.

3.3 ATM Uptime Monitoring Module

- **FR-UM-01: Downtime Recording**
 - The system shall provide an interface for Branch IT Officers to log an outage with: Start Time, End Time, Reason Category (Power, Network, Hardware, Cash-Out, Software), and detailed notes.
- **FR-UM-02: Uptime Calculation**
 - The system shall automatically calculate:
 - $\text{Daily Uptime \%} = [(24 - \text{Downtime in Hours}) / 24] * 100$
 - $\text{Monthly Uptime \%} = [(\text{Total Monthly Hours} - \text{Total Downtime}) / \text{Total Monthly Hours}] * 100$
- **FR-UM-03: Performance Dashboard**
 - The system shall display a dedicated dashboard with widgets for: Real-time Network Uptime %, Top cause of failures, All districts, and SLA Compliance Trends.

3.4 Handover and Takeover Module

- **FR-HT-01: Custody Transfer**
 - The system shall require the releasing officer ("Sender") to initiate a handover, specifying the equipment and its current condition. The receiving officer ("Receiver") must log in and explicitly acknowledge acceptance, creating an immutable custody record.

3.5 Reporting Module

- **FR-RP-01: Report Generation**
 - The system shall provide pre-configured reports (Inventory, Maintenance Summary, SLA Performance, Uptime, Branch Ranking, Audit Log) with filters for

Date Range, Branch, Equipment Type, and Status. Reports shall be exportable to Excel (.xlsx) and PDF formats.

3.6 User and Access Management

- **FR-UA-01: Authentication**
 - The system shall require username/password login, enforce password complexity, hash passwords using bcrypt, lock accounts after 5 failed attempts, and terminate sessions after 30 minutes of inactivity.
- **FR-UA-02: Authorization (RBAC)**
 - The system shall implement role-based access control. Permissions for Create, Read, Update, Delete, and Approve shall be granularly assignable to the roles defined in Section 2.3.

3.7 Audit Logging

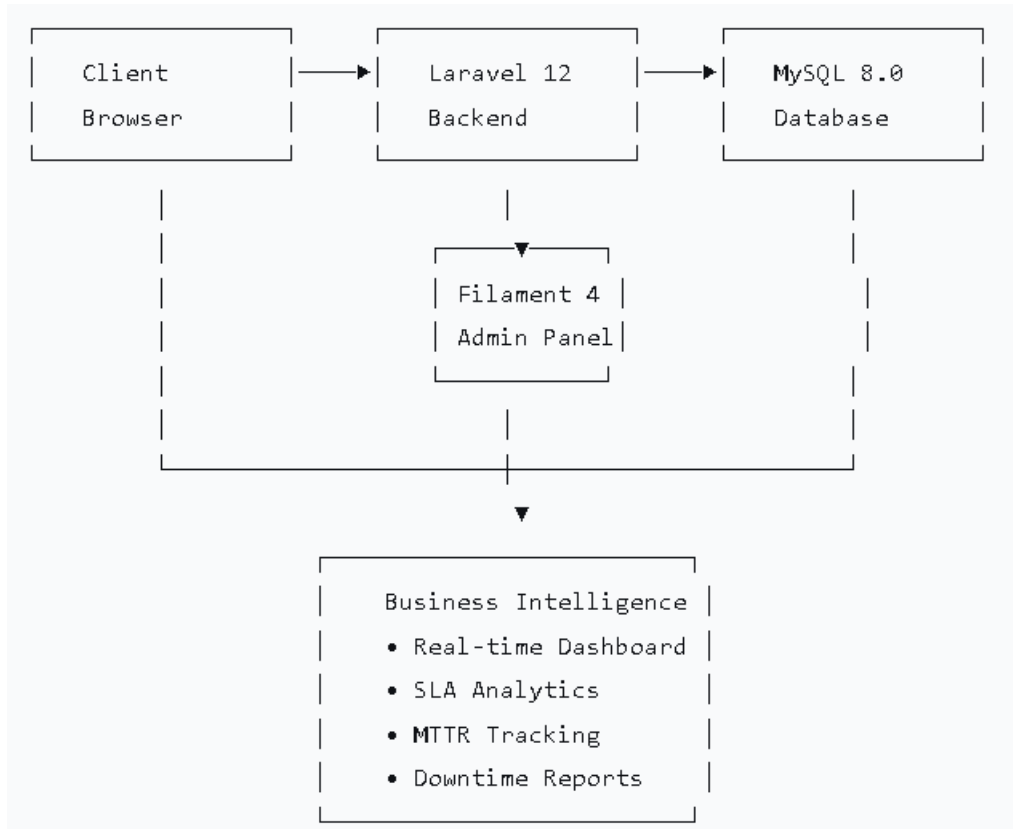
- **FR-AL-01: Comprehensive Logging**
 - The system shall automatically log all critical events: User Login/Logout, CRUD operations on all major entities (Equipment, Tickets, and Handovers), approval actions, and system changes. Each log entry shall include User ID, Action, IP Address, Timestamp, and relevant record ID.

3.2 Non-Functional Requirements

- **NF-PF-01: Performance**
 - All interactive web pages shall load within **3 seconds** under normal load.
 - Standard reports shall generate within **10 seconds**.
 - The system shall support **500 concurrent users** without significant performance degradation.

- **NF-SC-01: Security**
 - The system shall be protected against OWASP Top 10 vulnerabilities (SQL Injection, XSS, CSRF, etc.).
 - All sensitive data at rest (backups) and in transit shall be encrypted.
- **NF-AV-01: Availability**
 - The system shall achieve **99% operational uptime** during business hours (8 AM - 8 PM, Mon-Sat).
 - Full database backups shall be performed daily.
- **NF-SC-02: Scalability**
 - The database schema shall efficiently support **10,000+ equipment records** and **10 years** of transactional history.
- **NF-MT-01: Maintainability**
 - The codebase shall follow a modular, service-oriented architecture (e.g., MVC, Repository Pattern) to allow for easy future enhancements and bug fixes.

4. Technical Architecture



5. RISK ANALYSIS

Risk	Probability	Impact	Mitigation Strategy
Inaccurate manual downtime entry	Medium	High	Implement mandatory fields, reason codes, and supervisor approval for long outages.
SLA timer manipulation by users	Low	High	Make SLA timers system-calculated based on status changes; all changes logged immutably.
Data loss due to server failure	Low	Critical	Implement daily automated backups with off-site replication and a documented DR restore procedure.
User resistance to new workflow	High	Medium	Conduct extensive UAT, involve super-users in design, and provide clear training & support.

5. FUTURE ENHANCEMENTS

- **Phase 2:** Integrate with existing **ATM monitoring APIs** for real-time, automated status updates.
 - **Phase 2:** Implement **SMS/Email Alert System** for SLA breaches and critical downtime.
 - **Phase 3:** Develop a **progressive web app (PWA)** or native mobile app for field technicians.
 - **Roadmap:** Explore **predictive analytics** using historical MTBF/MTTR data to forecast failures
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APPENDICES

- **Appendix A:** Wireframes / UI Mockups (to be provided separately).
 - **Appendix B:** Detailed Data Dictionary / Database Schema.
 - **Appendix C:** Sample Report Formats.
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Document Approval:

Name & Title	Signature	Date
Admin Head	_____	
Manager of IT Operations	_____	
IT Infrastructure Director	_____	